

10 advanced-level Bubble.io assignment questions, each with **detailed feature requirements** designed to challenge learners who have already mastered intermediate workflows, dynamic content, responsive layouts, and plugin integrations.

Advanced-Level Bubble.io Assignments (10 Tasks)

Each assignment focuses on *complex logic, scalable architecture, performance optimization, or advanced integrations*.

1 Assignment 1 — Build a Multi-Tenant SaaS Platform

Objective: Learn data isolation, tenant-aware UX, and role-based access.

Features to Implement:

- Create **Organization** data type; users belong to an organization.
- Restrict data visibility so each organization sees only its own records.
- Implement **role levels**: Owner, Admin, Member.
- Add an onboarding flow where the Organization Owner sets up team members.
- Build a dashboard that updates based on the logged-in user's organization ID.
- Add usage-based metrics displayed in real-time (e.g., number of projects created).

2 Assignment 2 — Build a Real-Time Collaborative Editor

Objective: Use WebSockets/plugins, auto-refresh logic, and conflict resolution.

Features to Implement:

- Create a collaborative **Notes** or **Document** editor.
- Use an auto-updating repeating group or WebSocket plugin for real-time syncing.
- Implement **“user is typing” indicators**.
- Handle edit conflicts using timestamps or “last write wins.”
- Track each user's cursor position or active editing block.

3 Assignment 3 — Create an AI-Powered Feature Using External AI APIs

Objective: Integrate OpenAI or similar LLM APIs into Bubble workflows.

Features to Implement:

- Build a text input where users ask questions or submit content.
- API call returns AI-generated text (summary, rewrite, answer, analysis).
- Store AI responses in a dedicated “AI Log” data type.
- Allow user to refine the prompt and regenerate responses.
- Add per-user API usage tracking (count or billing).

4 Assignment 4 — Build a Complex Booking + Resource Management System

Objective: Manage overlapping events, availability logic, and notifications.

Features to Implement:

- Data types: Resource, Booking, TimeSlot, User.
- Allow resources (rooms, equipment, people) to have unique availability windows.
- Prevent **overlapping bookings** using complex constraints.
- Add color-coded calendar view (via plugin).
- Automated reminders using backend workflows (e.g., send reminder 2 hrs before).

5 Assignment 5 — Implement a Fully Custom E-Commerce Engine

Objective: Build a production-level store with checkout workflows.

Features to Implement:

- Product variants (size, color), inventory tracking, and dynamic pricing.
- Shopping cart stored per user (session-based for logged-out visitors).
- Integrate Stripe or Braintree (via plugin) for secure payments.
- Generate dynamic order confirmation PDFs sent via Postmark or email plugin.
- Build admin analytics for sales, refunds, and inventory.

6 Assignment 6 — Create a Social Network with Activity Feed Algorithms

Objective: Build scalable feed logic and multi-relation structures.

Features to Implement:

- Data types: Post, Comment, Reaction, Follow, Notification.
- Implement a **personalized feed algorithm** using:
 - Following relationships
 - Recent interactions
 - Trending posts (based on engagement score)
- Add lazy loading for infinite scrolling.
- Implement privacy rules for followers-only content.
- Add content moderation tools for admins.

7 Assignment 7 — Build a Custom API Layer (Bubble as Backend-Only)

Objective: Expose Bubble as a REST API for external apps.

Features to Implement:

- Use **API Workflow Endpoints** with authentication.
- Implement endpoints:
 - `POST /login`
 - `GET /items`
 - `POST /items`
 - `GET /items/:id`
- Configure rate limiting and API keys.
- Add logging of all external API calls to an "APIAudit" data type.
- Provide documentation inside the app using a dynamic documentation page.

8 Assignment 8 — Build a Custom Reporting & Data Visualization Engine

Objective: Advanced search constraints, aggregations, and dynamic filters.

Features to Implement:

- Build a **query builder UI** that allows:
 - Date range filters
 - Multi-field filtering (category, tag, user, status)
 - Sorting
- Generate live charts (bar, line, pie) using a chart plugin.
- Add export options: CSV, Excel, and PDF reports.
- Build a "Saved Reports" feature stored per user.

9 Assignment 9 — Build a Marketplace with Bidding or Auction Features

Objective: Real-time bidding, countdown timers, and event triggers.

Features to Implement:

- Data types: AuctionItem, Bid, User, AuctionEventLogs.
- Show highest bid in real-time (auto refresh or WebSocket plugin).
- Implement countdown timers for auction closing.
- When auction ends:
 - Lock item
 - Create “WinningBid” record
 - Trigger backend workflow for notifications
- Create an admin dashboard for managing auctions.

10 Assignment 10 — Create a Full CRM System with Pipelines & Automation

Objective: Pipeline UI, automation rules, and backend logic.

Features to Implement:

- Data types: Lead, Contact, Deal, Stage, Task.
- Visual **Kanban board** to drag deals across pipeline stages.
- Automatic rule engine for:
 - Send email when deal enters Stage X
 - Assign task to team member
 - Score leads based on activity
- Build analytics dashboard (deal conversion %, revenue projections).
- Add role-based access and team-level visibility.